

Chapter Notes 2

Day 2: Covariance & Correlation Matrices

STAT 4051 — Statistical Machine Learning I | University of Minnesota

Topic	Covariance & Correlation Matrices
Week	Week 1 — Foundations: Multivariate Statistics & PCA
Prerequisites	Day 1 (data matrix $\mathbf{X}$, mean vector $\bar{\mathbf{x}}$, centering)
Key Ideas	$\mathbf{S}$, $\mathbf{R}$, PSD matrices, geometric interpretation, standardization
Connects To	Day 3: Eigendecomposition & SVD; Days 4–5: PCA

Learning Objectives.

- Derive the sample covariance matrix $\mathbf{S}$ from first principles and express it in matrix form.
- Identify and prove the key properties of $\mathbf{S}$: symmetry, positive semi-definiteness, and units.
- Construct the sample correlation matrix $\mathbf{R}$ via standardization and understand its relationship to $\mathbf{S}$.
- Interpret covariance and correlation geometrically as inner products and cosines of angles.
- Distinguish the population covariance matrix Σ from its sample estimator $\mathbf{S}$.
- Recognize the spectral constraints on $\mathbf{S}$ that motivate PCA on Day 4.

From Marginals to Relationships

On Day 1 we summarized each variable separately: means, variances, histograms. These *marginal* summaries deliberately ignore the relationships between variables. But in multivariate data, those relationships carry the most interesting information.

A covariance matrix encodes, in a single $p \times p$ array, how every pair of variables moves together across observations. It is the central object in multivariate statistics: PCA decomposes it, factor analysis models it, and Gaussian graphical models invert it. Everything in Weeks 1–3 of this course traces back to understanding $\mathbf{S}$.

The Sample Covariance Matrix

Scalar Definition

Recall: the sample variance of a single variable j is

$$s_j^2 = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2.$$

The *sample covariance* between variables j and k generalizes this by replacing the squared deviation with a product of two centered deviations:

$$s_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k).$$

Notice $s_{jj} = s_j^2$ (variance is a special case of covariance). Also $s_{jk} = s_{kj}$, so the matrix we are about to build is symmetric.

Definition 2.1 (Sample Covariance Matrix). *The sample covariance matrix is the $p \times p$ symmetric matrix*

$$\mathbf{S} = [s_{jk}]_{j,k=1}^p = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top,$$

where $\mathbf{x}_i \in \mathbb{R}^p$ is the i -th observation (row i of $\mathbf{X}$, written as a column vector) and $\bar{\mathbf{x}} \in \mathbb{R}^p$ is the sample mean vector.

The outer product $(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$ is a $p \times p$ rank-1 matrix capturing how observation i 's deviation from the mean spreads across all pairs of variables. Summing n such rank-1 matrices and dividing by $n-1$ gives $\mathbf{S}$.

Matrix Form

Working observation-by-observation is inefficient. Let $\tilde{\mathbf{X}}$ denote the *mean-centered data matrix*:

$$\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{1}_n \bar{\mathbf{x}}^\top \in \mathbb{R}^{n \times p},$$

where $\mathbf{1}_n = (1, 1, \dots, 1)^\top \in \mathbb{R}^n$ and each row of $\mathbf{1}_n \bar{\mathbf{x}}^\top$ is $\bar{\mathbf{x}}^\top$. Row i of $\tilde{\mathbf{X}}$ is $\mathbf{x}_i - \bar{\mathbf{x}}$, the centered observation vector.

Proposition 2.1 (Matrix form of $\mathbf{S}$).

$$\mathbf{S} = \frac{1}{n-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}.$$

Proof. The (j, k) entry of $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$ is the dot product of column j and column k of $\tilde{\mathbf{X}}$:

$$(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})_{jk} = \sum_{i=1}^n \tilde{x}_{ij} \tilde{x}_{ik} = \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k).$$

Dividing by $n-1$ gives s_{jk} . □ □

Why $n-1$ and not n ? Dividing by $n-1$ (Bessel's correction) makes $\mathbf{S}$ an *unbiased* estimator of the population covariance matrix Σ : $\mathbb{E}[\mathbf{S}] = \Sigma$. Dividing by n gives the MLE under a Gaussian model, which is biased but has lower MSE. R's `cov()` uses $n-1$ by default; most ML libraries use n . Always check which convention is in use.

An Equivalent Expression via the Centering Matrix

The centered matrix $\tilde{\mathbf{X}}$ can be written as $\tilde{\mathbf{X}} = \mathbf{H}\mathbf{X}$, where

$$\mathbf{H} = \mathbf{I}_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top$$

is the $n \times n$ *centering matrix* (also called the hat matrix's complement). $\mathbf{H}$ is symmetric and idempotent ($\mathbf{H}^2 = \mathbf{H}$), and it projects any vector onto the subspace orthogonal to $\mathbf{1}_n$. This gives:

$$\mathbf{S} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{H}^\top \mathbf{H} \mathbf{X} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{H} \mathbf{X}.$$

You will encounter $\mathbf{H}$ again in regression (where it projects onto the column space of the design matrix) and in kernel methods.

Properties of the Sample Covariance Matrix

Symmetry

$\mathbf{S} = \mathbf{S}^\top$ follows immediately from $s_{jk} = s_{kj}$. Equivalently, $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^\top = \tilde{\mathbf{X}}^\top (\tilde{\mathbf{X}}^\top)^\top = \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$.

Positive Semi-Definiteness

Proposition 3.1. $\mathbf{S}$ is positive semi-definite (PSD): for all $\mathbf{v} \in \mathbb{R}^p$, $\mathbf{v}^\top \mathbf{S} \mathbf{v} \geq 0$.

Proof. Let $\mathbf{u} = \tilde{\mathbf{X}} \mathbf{v} \in \mathbb{R}^n$. Then

$$\mathbf{v}^\top \mathbf{S} \mathbf{v} = \frac{1}{n-1} \mathbf{v}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{v} = \frac{1}{n-1} \|\mathbf{u}\|^2 = \frac{1}{n-1} \|\tilde{\mathbf{X}} \mathbf{v}\|^2 \geq 0. \quad \square$$

□

The quadratic form $\mathbf{v}^\top \mathbf{S} \mathbf{v}$ has a clean interpretation: it is (up to scaling) the sample variance of the *projected* variable $\tilde{\mathbf{X}} \mathbf{v}$, i.e., of the linear combination $v_1 x_1 + \dots + v_p x_p$ across all observations. Since variances are non-negative, $\mathbf{S}$ must be PSD.

PSD implies all eigenvalues of $\mathbf{S}$ are non-negative (we will prove this rigorously on Day 3). This is the fact PCA exploits: the eigenvalues of $\mathbf{S}$ measure the variance explained by each principal direction.

$\mathbf{S}$ is PSD but generally not positive *definite* (PD). It is PD only when $\tilde{\mathbf{X}}$ has full column rank, i.e., when $\text{rank}(\tilde{\mathbf{X}}) = p$. This requires $n > p$ and no exact linear dependencies among variables. In high-dimensional settings where $p > n$, $\mathbf{S}$ is rank-deficient, and naive inversion is impossible — a problem directly motivating methods like regularized covariance estimation and PCA.

Units and Diagonal Entries

The (j, j) entry s_{jj} is the sample variance of variable j , with units of $(\text{unit of } x_j)^2$. The off-diagonal entry s_{jk} has units of $(\text{unit of } x_j) \times (\text{unit of } x_k)$. This means $\mathbf{S}$ is not invariant under changes of measurement scale: multiply variable j by a constant c and s_{jk} scales by c . This scale-dependence motivates standardization.

The Sample Correlation Matrix

Standardization

To remove the effect of measurement scale, replace each variable j with its *standardized* version:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j},$$

where $s_j = \sqrt{s_{jj}}$ is the sample standard deviation of variable j . The standardized variable has sample mean 0 and sample variance 1 by construction.

Let $\mathbf{Z} \in \mathbb{R}^{n \times p}$ denote the matrix of all standardized values, and define

$$\mathbf{D} = \text{diag}(s_1, s_2, \dots, s_p) \in \mathbb{R}^{p \times p}.$$

Then column j of $\mathbf{Z}$ is column j of $\tilde{\mathbf{X}}$ divided by s_j , so $\mathbf{Z} = \tilde{\mathbf{X}} \mathbf{D}^{-1}$.

Definition 4.1 (Sample Correlation Matrix). *The sample correlation matrix is*

$$\mathbf{R} = \frac{1}{n-1} \mathbf{Z}^\top \mathbf{Z} = \mathbf{D}^{-1} \mathbf{S} \mathbf{D}^{-1}.$$

Its (j, k) entry is the sample Pearson correlation coefficient:

$$r_{jk} = \frac{s_{jk}}{s_j s_k} = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2} \sqrt{\sum_{i=1}^n (x_{ik} - \bar{x}_k)^2}}.$$

Properties of $\mathbf{R}$

- **Symmetry:** $\mathbf{R} = \mathbf{R}^\top$ (inherited from $\mathbf{S}$).
- **Diagonal entries:** $r_{jj} = s_{jj}/s_j^2 = 1$ for all j . The diagonal of $\mathbf{R}$ is all ones.
- **Bounded entries:** $r_{jk} \in [-1, 1]$ for all j, k . This is the Cauchy-Schwarz inequality applied to the centered columns of $\mathbf{X}$.
- **PSD:** $\mathbf{R}$ is PSD for the same reason $\mathbf{S}$ is: $\mathbf{R} = \mathbf{D}^{-1} \mathbf{S} \mathbf{D}^{-1}$, and pre- and post-multiplying a PSD matrix by any invertible matrix preserves PSD-ness.
- **Scale-invariance:** $\mathbf{R}$ is unchanged if any variable is multiplied by a positive constant (the constant cancels in the numerator and denominator of r_{jk}).

Relationship: $\mathbf{S}$, $\mathbf{R}$, and $\mathbf{D}$

The three objects are linked by:

$$\mathbf{S} = \mathbf{D} \mathbf{R} \mathbf{D}, \quad \mathbf{R} = \mathbf{D}^{-1} \mathbf{S} \mathbf{D}^{-1}.$$

Knowing any two of $\{\mathbf{S}, \mathbf{R}, \mathbf{D}\}$ determines the third. In practice:

- Use $\mathbf{S}$ when variable scales are meaningful and comparable (e.g., all variables in the same units).
- Use $\mathbf{R}$ (equivalently, standardize $\mathbf{X}$ first) when variables are on different scales or when you do not want PCA or clustering to be dominated by high-variance variables.

Geometric Interpretation

Covariance as an Inner Product

Let $\tilde{\mathbf{x}}^{(j)} \in \mathbb{R}^n$ denote column j of $\tilde{\mathbf{X}}$ (the centered j -th variable). Then:

$$s_{jk} = \frac{1}{n-1} \langle \tilde{\mathbf{x}}^{(j)}, \tilde{\mathbf{x}}^{(k)} \rangle = \frac{1}{n-1} \|\tilde{\mathbf{x}}^{(j)}\| \|\tilde{\mathbf{x}}^{(k)}\| \cos \theta_{jk},$$

where θ_{jk} is the angle between the two centered column vectors in $\mathbb{R}^n$. Dividing by the norms gives:

$$r_{jk} = \cos \theta_{jk}.$$

The sample correlation between variables j and k is the cosine of the angle between their mean-centered column vectors. This gives immediate intuition:

- $r_{jk} = 1$: vectors point in exactly the same direction — perfect positive linear relationship.
- $r_{jk} = -1$: vectors point in exactly opposite directions — perfect negative linear relationship.
- $r_{jk} = 0$: vectors are orthogonal — no linear relationship.

This geometric view is the conceptual foundation for PCA: finding the directions in $\mathbb{R}^p$ (observation space) that explain the most variance corresponds to finding the eigenvectors of $\mathbf{S}$.

Mahalanobis Distance

The inverse of the covariance matrix (when it exists) defines the *Mahalanobis distance* between two observations $\mathbf{x}_i$ and $\mathbf{x}_j$:

$$d_M(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^\top \mathbf{S}^{-1} (\mathbf{x}_i - \mathbf{x}_j)}.$$

This is a scale-invariant distance that accounts for the correlation structure of the data. It will reappear in discriminant analysis and the multivariate normal distribution.

Population vs. Sample

So far everything has been *sample* quantities computed from data. The corresponding *population* objects are defined in terms of expected values.

Definition 6.1 (Population Covariance Matrix). *For a random vector $\mathbf{X} \in \mathbb{R}^p$ with mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$, the population covariance matrix is*

$$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] \in \mathbb{R}^{p \times p}.$$

Entry (j, k) is $\sigma_{jk} = \text{Cov}(X_j, X_k)$.

The sample covariance matrix $\mathbf{S}$ estimates Σ . Under mild conditions (finite second moments, i.i.d. observations), the law of large numbers guarantees $\mathbf{S} \rightarrow \Sigma$ as $n \rightarrow \infty$.

The Multivariate Normal. If $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$, then Σ completely characterizes the dependence structure. The density is:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right).$$

The MLE of Σ given n i.i.d. observations is $\hat{\Sigma}_{MLE} = \frac{1}{n} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$ (divides by n , not $n - 1$). This is biased but has smaller mean squared error.

Spectral Preview: Why Eigenvalues Matter

Because $\mathbf{S}$ is real, symmetric, and PSD, the spectral theorem guarantees it can be written as

$$\mathbf{S} = \mathbf{V} \boldsymbol{\Lambda} \mathbf{V}^\top,$$

where $\mathbf{V} \in \mathbb{R}^{p \times p}$ is orthogonal (columns are eigenvectors) and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_p)$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ (eigenvalues, all non-negative).

Key quantity: The *total variance* of the dataset is

$$\text{tr}(\mathbf{S}) = \sum_{j=1}^p s_{jj} = \sum_{j=1}^p \lambda_j.$$

The fraction of total variance explained by the k -th principal component is $\lambda_k / \sum_j \lambda_j$. We will derive this carefully on Day 4; for now, internalize that the eigenvalues of $\mathbf{S}$ are the fundamental measure of how much information each direction carries.

Worked Example: The iris Dataset

The `iris` dataset has $n = 150$ observations and $p = 4$ numeric variables (sepal length, sepal width, petal length, petal width), all in centimeters.

Mean Vector

$$\bar{\mathbf{x}} = \begin{pmatrix} 5.843 \\ 3.057 \\ 3.758 \\ 1.199 \end{pmatrix} \quad (\text{all in cm}).$$

Covariance Matrix

$$\mathbf{S} = \begin{pmatrix} 0.686 & -0.042 & 1.274 & 0.516 \\ -0.042 & 0.190 & -0.330 & -0.122 \\ 1.274 & -0.330 & 3.116 & 1.296 \\ 0.516 & -0.122 & 1.296 & 0.581 \end{pmatrix}$$

Observations: the diagonal entries (variances) are largest for petal length (3.116); petal length and petal width are strongly positively covariant ($s_{34} = 1.296$); sepal width has a slight negative covariance with all other variables.

Correlation Matrix

$$\mathbf{R} = \begin{pmatrix} 1.000 & -0.118 & 0.872 & 0.818 \\ -0.118 & 1.000 & -0.428 & -0.366 \\ 0.872 & -0.428 & 1.000 & 0.963 \\ 0.818 & -0.366 & 0.963 & 1.000 \end{pmatrix}$$

Observations: petal length and petal width are nearly perfectly correlated ($r = 0.963$), suggesting they carry nearly redundant information. Sepal width is weakly (and negatively) correlated with the other three variables.

Total variance: $\text{tr}(\mathbf{S}) = 0.686 + 0.190 + 3.116 + 0.581 = 4.573 \text{ cm}^2$. Day 4 will show that the first principal component explains about 73% of this total.

Summary

Object	Formula	Key Property
Sample covariance	$\mathbf{S} = \frac{1}{n-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$	Symmetric, PSD, units of x^2
Sample correlation	$\mathbf{R} = \mathbf{D}^{-1} \mathbf{S} \mathbf{D}^{-1}$	Symmetric, PSD, diagonal = 1, entries in $[-1, 1]$
Standardized data	$\mathbf{Z} = \tilde{\mathbf{X}} \mathbf{D}^{-1}$	$\text{cov}(\mathbf{Z}) = \mathbf{R}$
Centering matrix	$\mathbf{H} = \mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top$	Symmetric, idempotent, $\mathbf{S} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{H} \mathbf{X}$
Mahalanobis dist.	$d_M = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^\top \mathbf{S}^{-1} (\mathbf{x}_i - \mathbf{x}_j)}$	Scale-invariant, accounts for correlation

What's next. Day 3 derives the eigendecomposition and SVD of $\mathbf{S}$ from scratch. Every spectral result stated without proof today will be proved rigorously there.